



Mecanum-wheeled Autonomous Tracking Robot Thesis Proposal

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Trace Rover Team



Contract

1. Project Background and Topic Selection
2. Structure
3. Control
4. Electronic
5. Vision
6. Software
7. Budget
8. Summary

Project Background and Topic Selection

Market Background & Industry Needs

The demand for agile maneuvering has surged, while traditional chassis have encountered bottlenecks.

Macro Demand: In complex scenarios such as intelligent manufacturing, warehouse logistics, and disaster area reconnaissance, robots need to shuttle and track targets with high precision in narrow, unstructured spaces.

Existing problems: Traditional differential-drive chassis suffer from large turning radius and limited maneuverability. Existing omnidirectional Mecanum-wheel chassis improve mobility, but severe vibration and slippage often occur in practice due to rigid contact.

Industry Needs: There is an urgent need for a mobile platform with both extreme **maneuverability** and high **all-terrain stability** to serve as a reliable carrier for visual tracking and robotic arm operation.



Literature Review & Technical Pain Points

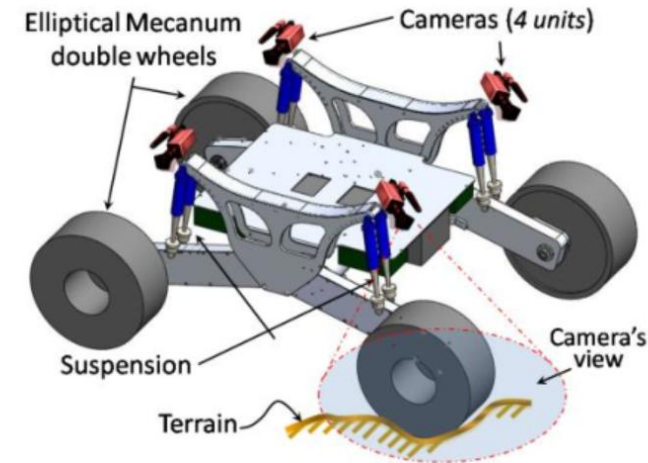
Technical Pain Points

Chassis Vibration and Lack of Suspension:

Intermittent contact of Mecanum wheel rollers causes periodic vibration, and systems without independent suspension lose grip on uneven roads (Ref. 1, 2).

Nonlinear Slip and Trajectory Drift: Skidding makes ideal kinematic matrices ineffective for accurate dead reckoning, and traditional single closed-loop PID easily leads to trajectory deviation in high-speed response (Ref. 3, 4).

Separation of Computing Power and Control: Fragmented "upper computer + lower controller" solutions lack bottom-level high-density integration, resulting in system delay between sensor fusion and motor response (Ref. 5).



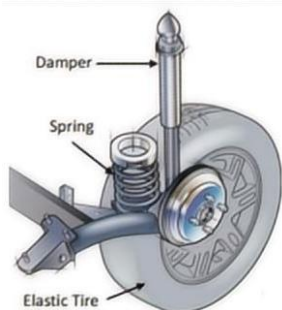
Reference

1. Oh, S. et al. (2021). "Vibration Reduction Control of Omnidirectional Mobile Robot with Lift Mechanism." IEEE International Conference on Control, Automation and Systems (ICCAS).
2. Chen, Y. et al. (2024). "Systematic Review of Mecanum and Omni Wheel Technologies for Motor Impairments." MDPI Applied Sciences.
3. Wang, H. et al. (2019). "Trajectory tracking for omnidirectional mecanum robot with longitudinal slipping." MATEC Web of Conferences.
4. Li, X. et al. (2018). "Highly Maneuverable Ground Reconnaissance Robot Based on Machine Learning." IEEE International Conference on Robotics and Biomimetics (ROBIO).
5. Zhang, M. et al. (2020). "Design of Inspection Robot for Spherical Tank Based on Mecanum Wheel." IEEE International Conference on Mechatronics and Automation (ICMA).

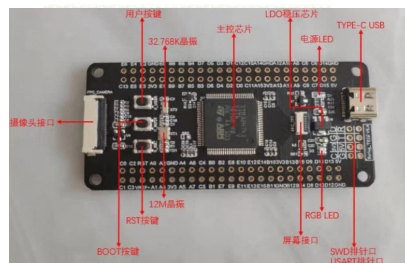
Our Work

We are not "assembling" a small car, but building a highly coupled complex robot system from scratch.

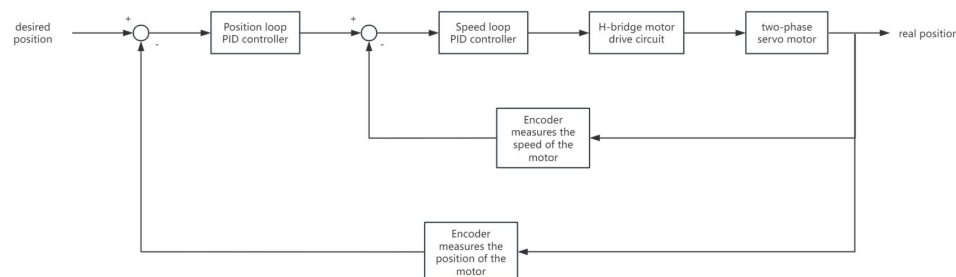
System Closed-Loop:



Mechanical Layer: Passive compliance for road adaptability via independent spring-damper suspension.



Hardware Layer: STM32H7 550MHz core for extreme space compression and ultra-low latency at the bottom board level.



Control & Perception Layer: Cascade PID motor closed-loop + encoder dead reckoning + ultrasonic polar radar + YOLO visual tracking. Data of each subsystem is deeply fused at high frequency, with every part affecting the whole system.

Project Background and Topic Selection

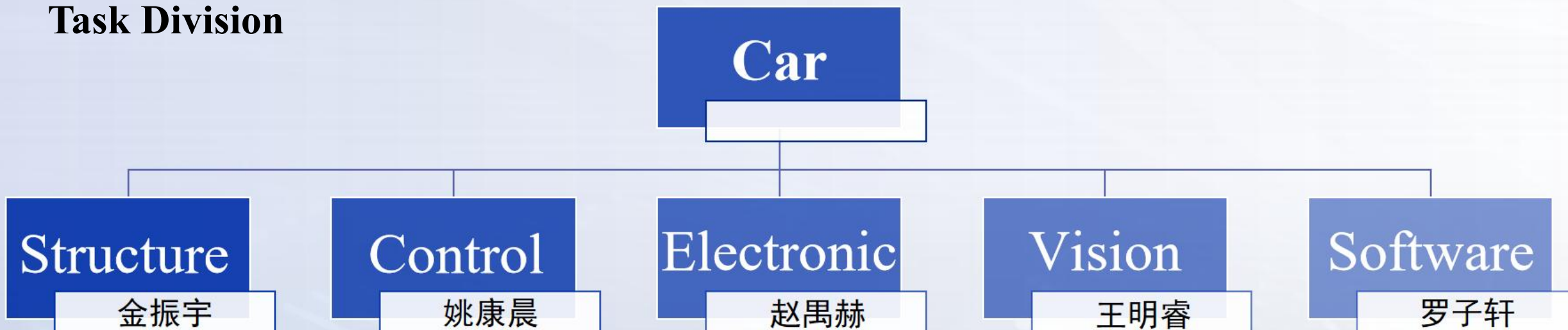
Objectives

Design and verify a highly stable omnidirectional mobile platform with a modular hierarchical architecture, low center of gravity, and a 2-DOF tracking gimbal.

Challenges

1. Parameter Matching and Tuning of Independent Suspension System in Narrow Space.
2. Multi-Motor Cooperative Closed-Loop and Dead Reckoning Error Handling of Mecanum Wheel Chassis.
3. Efficient Data Communication and Task Scheduling of Heterogeneous Software and Hardware Systems.

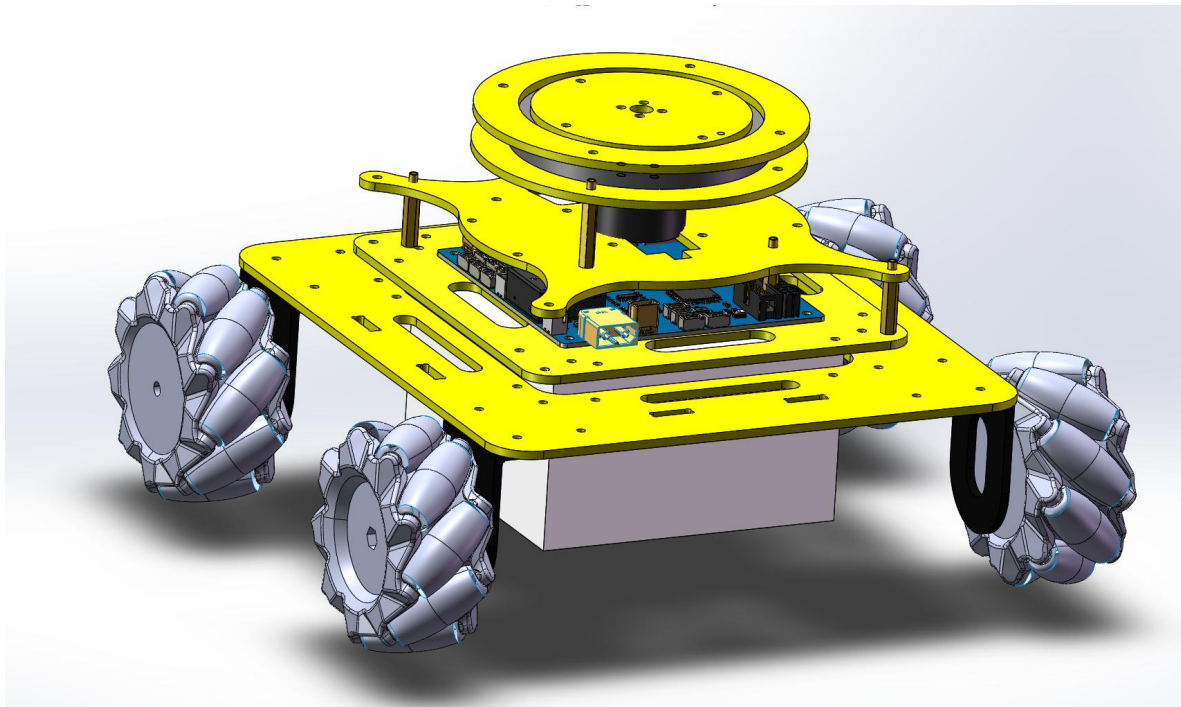
Task Division



Core objective

Design and build a modular, high-stability, lightweight Mecanum-wheel mobile platform integrated with a 2-DOF gimbal to provide a reliable motion and execution carrier for tracking robots.

Overall Structural Design



Modular Layered Architecture:

Segregates the chassis, control circuits, and tracking pan-tilt layers.

Ensures streamlined hardware debugging and scalable future expansion.

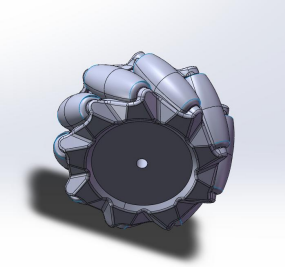
Optimized Spatial Layout:

Features a low center of gravity by placing batteries and heavy masses at the base, maximizing kinematic stability.

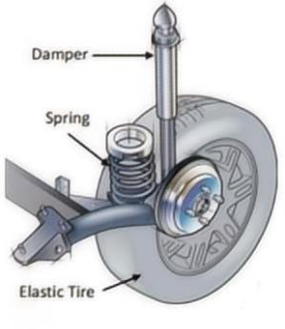
Ample upper-layer spacing ensures optimal heat dissipation for the control boards.

Core Mechanical Modules

Omnidirectional Chassis

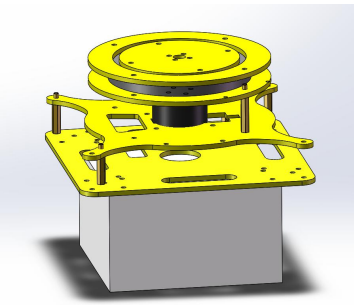


Adopt **Mecanum wheels** to achieve zero-radius turning and meet real-time tracking requirements.



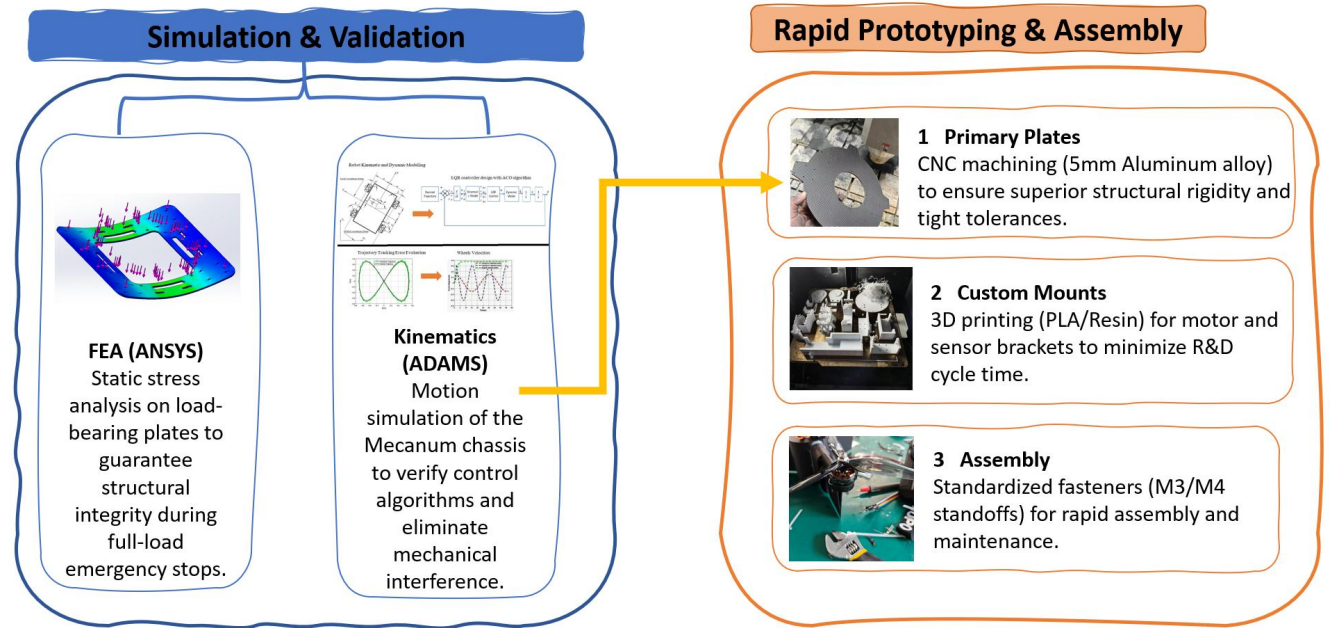
Design **independent spring-damper suspension** for each wheel to solve the problem of simultaneous four-wheel contact and improve adaptability on uneven roads.

2-DOF Tracking Pan-Tilt



Lightweight disc structure, driven by dual servos, achieves fast and omnidirectional target locking.

Simulation & Manufacturing



Theoretical modeling and parameter optimization

Dynamic Modeling: We model the suspension as a quarter-car linear spring-damper system. The equation of motion is defined as:

$$m\ddot{x} + c\dot{x} + kx = F_{road}(t)$$

(Where m is quarter-mass, c is damping coefficient, k is spring stiffness).

Parameter Optimization: To prevent chassis resonance and optimize target-tracking stability, we calculate the natural frequency ω_n and select springs to achieve an underdamped response:

$$\omega_n = \sqrt{\frac{k}{m}}, \quad \zeta = \frac{c}{2\sqrt{km}} < 1$$

Stiffness Selection: Based on a max payload of W_{max} and desired deflection δ_{max} , the required stiffness is determined by: $k = \frac{W_{max}}{4\delta_{max}}$.

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Subtasks	Specific Content
1.1 Modeling Design	3D modeling of chassis, suspension and gimbal (SolidWorks); material selection and preliminary layout
1.2 Simulation	Dynamic simulation of suspension system (e.g. ANSYS/Simulink); kinematic verification; parameter optimization
1.3 Manufacturing	CNC machining of main structural parts; 3D printing of brackets; procurement of standard parts; assembly and preliminary debugging

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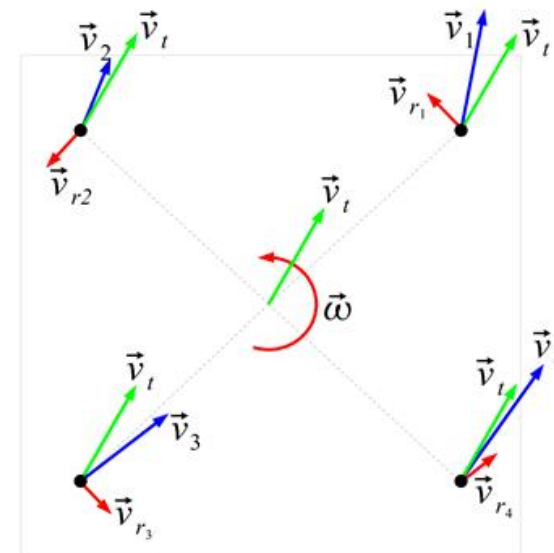
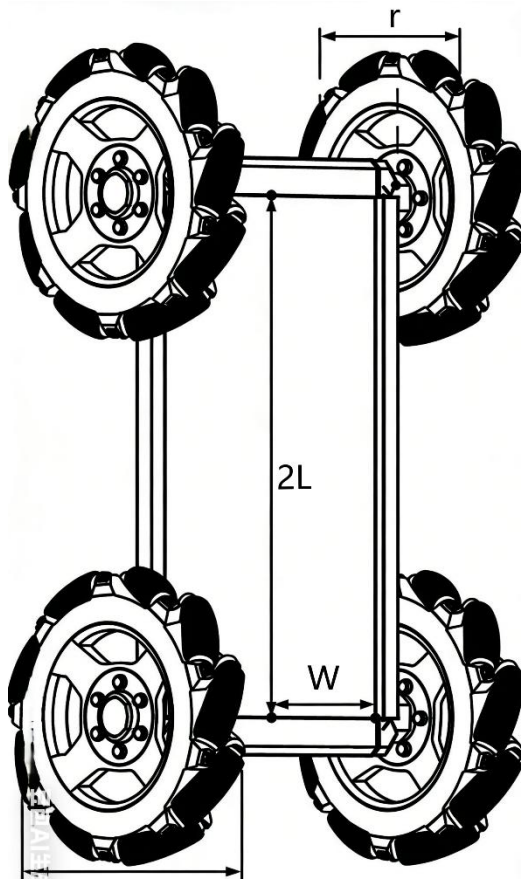
Motion control of Mecanum wheels

In order to control the movement of the AGV in the **3 degrees of freedom** on the plane, we need its

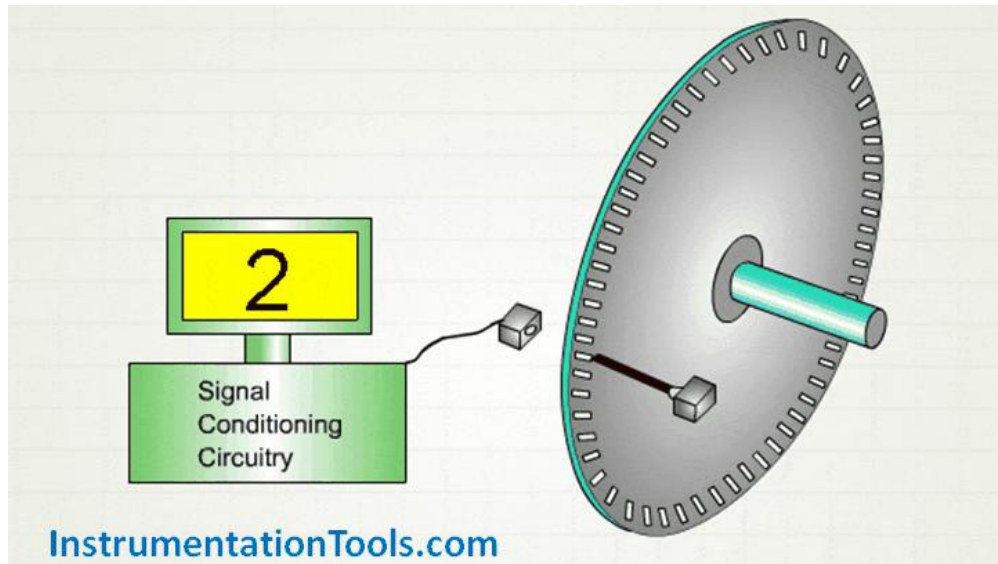
inverse kinematics matrix

$$\begin{bmatrix} \dot{s}_1 \\ \dot{s}_2 \\ \dot{s}_3 \\ \dot{s}_4 \end{bmatrix} = \begin{bmatrix} 1 & 1 & L + W \\ 1 & -1 & L + W \\ 1 & -1 & -(L + W) \\ 1 & 1 & -(L + W) \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$$

Through inverse kinematics, we can calculate the **required speeds/positions** for each motor by simply inputting the speed/displacement values in the **Cartesian coordinate system**.



Inertial navigation system based on encoder



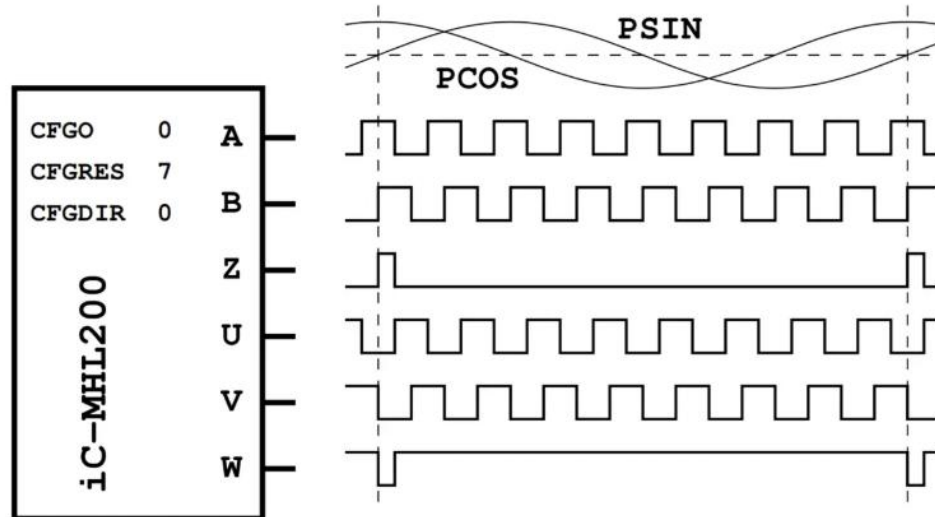
Encoders are used to translate **rotary or linear motion** into a **digital signal**. Usually this is for the purpose of monitoring or controlling motion parameters such as **speed, rate, direction, distance or position**.

After obtaining the **rotation distances** of the four wheels through the encoder, we can calculate the position and orientation of the AGV using the

forward kinematics matrix

$$\begin{bmatrix} x \\ y \\ \theta \end{bmatrix} = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & \frac{1}{4} \\ \frac{1}{4(L+W)} & -\frac{1}{4(L+W)} & \frac{1}{4(L+W)} & -\frac{1}{4(L+W)} \end{bmatrix} \begin{bmatrix} S_1 \\ S_2 \\ S_3 \\ S_4 \end{bmatrix}$$

In this way, we can calculate the AGV's position and orientation in real time using the encoder data from the four wheels.



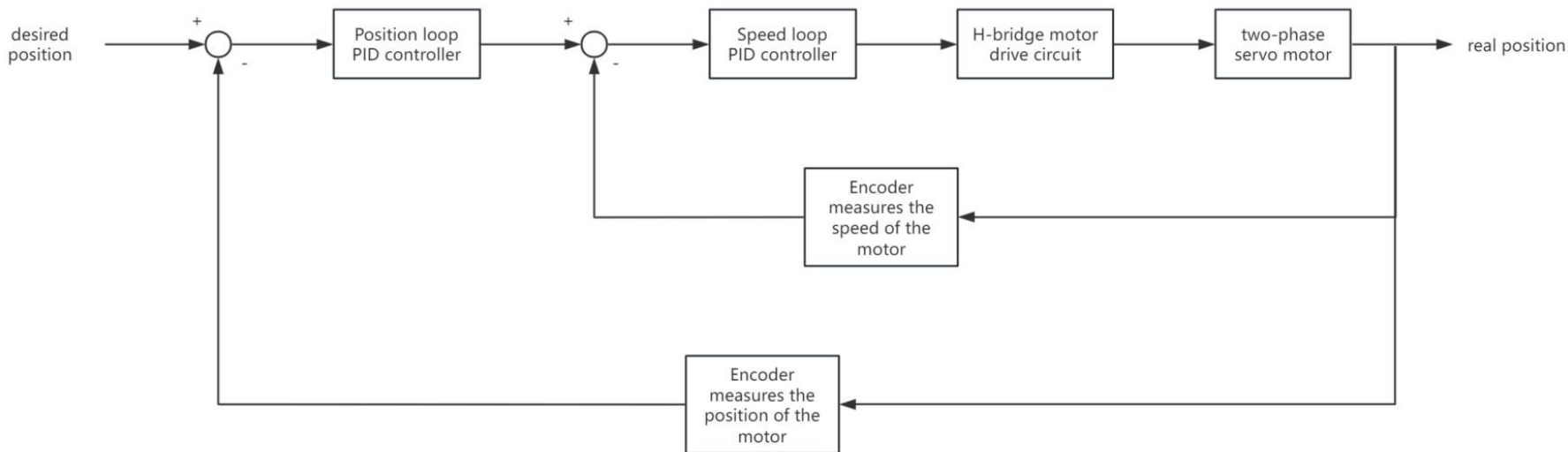
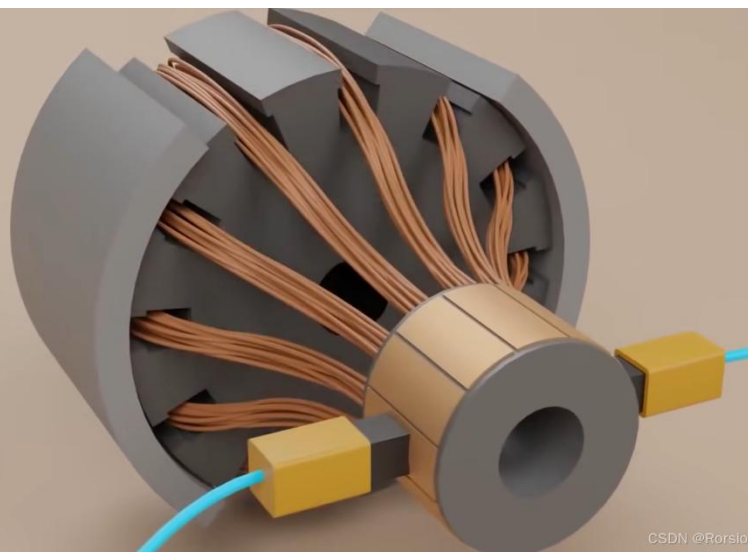
Closed-loop control of two-phase servo motors

The **transfer function** of the ideal two-phase servo motor system is

$$G(s) = \frac{\Omega_m(s)}{U_a(s)} = \frac{K_m}{T_m s + 1}$$

Although the ideal two-phase servo motor system is a simple first-order linear system, this system is prone to dynamic lag and parameter fluctuations due to the presence of inductance, inertia, and other parameters, and the position and speed variables are strongly coupled. A single-loop PID is unable to balance both dynamic response and positioning accuracy, so a **cascade PID** is designed. The **block diagram** is shown as follows.

After applying this controller, we can precisely adjust the position and speed of the motor. Combined with the previous **kinematic analysis** and inertial **navigation system**, we can freely control the **position** of the AGV.



An obstacle avoidance system composed of an ultrasonic sensor and a stepper motor

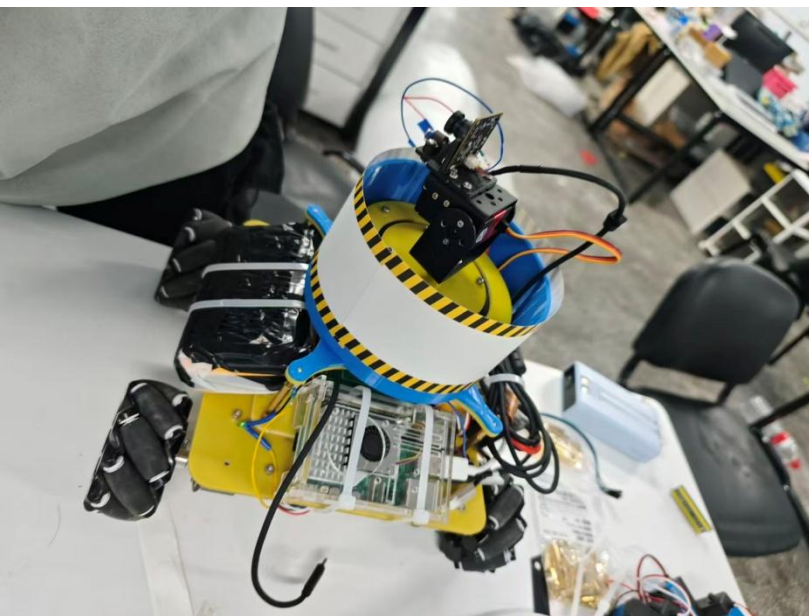
We use a **stepper motor** to drive the **ultrasonic sensor** installed on the **conductive slip ring** through a **gear mechanism**, causing it to **rotate continuously**. By combining the **angle information** from the stepper motor with the **distance information** input by the ultrasonic sensor, we can establish a **polar coordinate system** with the AGV as the origin to describe the information about the surrounding obstacles. Through the following algorithm, the obstacle information in the polar coordinate system can be converted into **Cartesian coordinates**.

$$\begin{cases} x_{obstacle} = R\cos(\phi_{obstacle}) \\ y_{obstacle} = R\sin(\phi_{obstacle}) \end{cases}$$

We can also utilize the **inertial navigation system** of the AGV to transform the obstacle information from the **robot's coordinate system** to the **world coordinate system**.

$$\begin{bmatrix} x_{world} \\ y_{world} \end{bmatrix} = \begin{bmatrix} \cos\theta_{robot} & -\sin\theta_{robot} \\ \sin\theta_{robot} & \cos\theta_{robot} \end{bmatrix} \begin{bmatrix} x_{obstacle} \\ y_{obstacle} \end{bmatrix} + \begin{bmatrix} x_{robot} \\ y_{robot} \end{bmatrix}$$

The host computer records all the obstacle points detected by the AGV. We can even implement a simple **SLAM** system.



Subtasks	Specific Content
2.1 Module Programming	Write the programs for each module, such as the filter module, the controller module, and the navigation system module program.
2.2 Module Testing	Test each module independently to ensure that there are no issues with the modules during the later system development process.
2.3 MCU System Design	Reorganize the previous modules into tasks within the RTOS, maintaining a system design with high cohesion and low coupling.

Project Start Date		3/13/2026 (Friday)		Display Week		1		Week 1							Week 2							Week 3							Week 4							Week 5							Week 6							Week 7							Week 8						
Project Lead								9 Mar 2026							16 Mar 2026							23 Mar 2026							30 Mar 2026							6 Apr 2026							13 Apr 2026							20 Apr 2026							27 Apr 2026						
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Core Objective

Design a **highly integrated, low latency, and highly reliable** electronic hardware platform for the autonomous tracking robot of the Mecanum wheel, achieving main control computation, multi-sensor fusion, and stable power supply, ensuring millisecond level response.

Design Concept

Integrated driver and control:

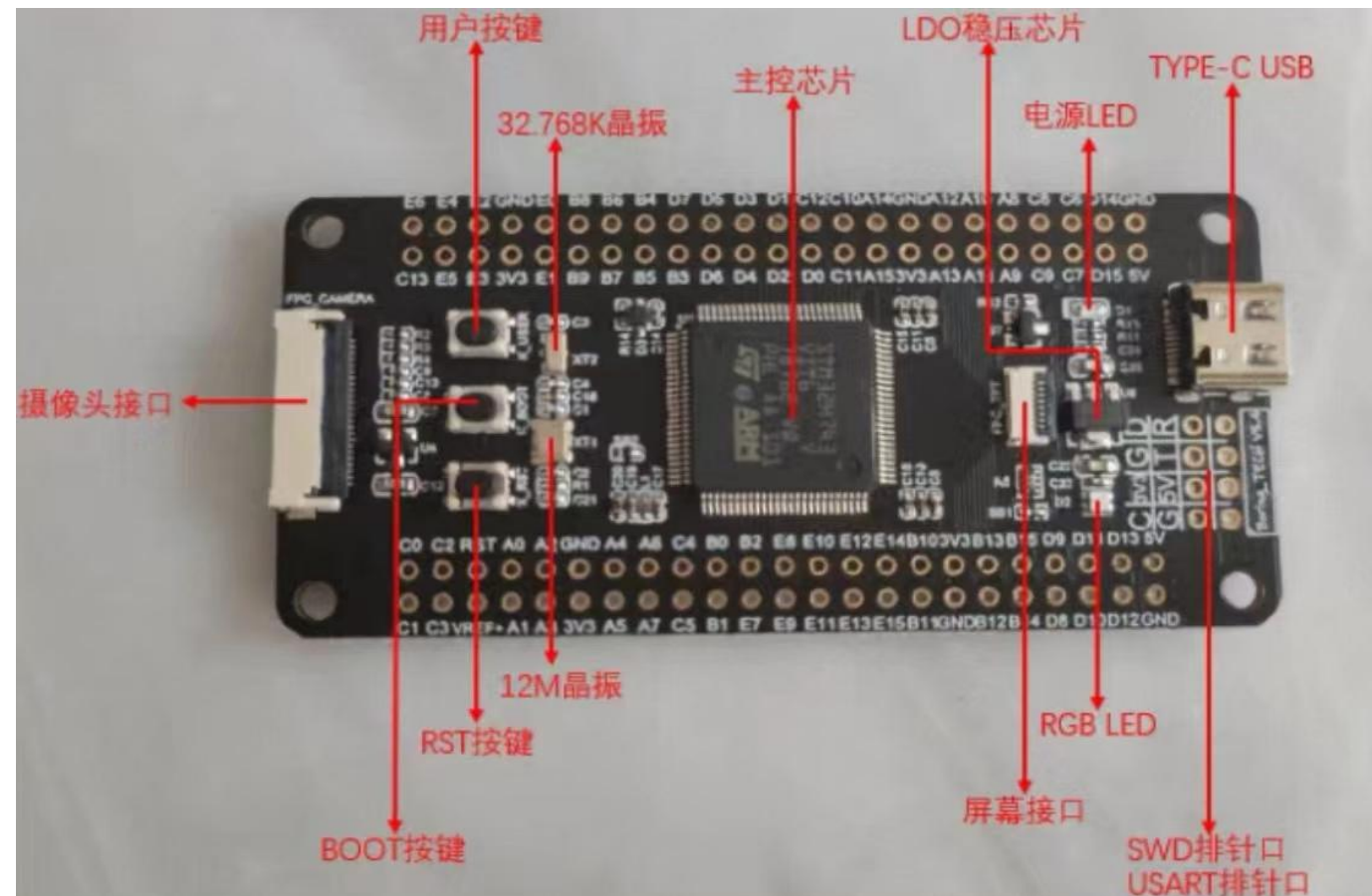
High-density integration of the main control MCU, power array, and peripheral interfaces on a single carrier board, compressing space and improving system compactness.

High performance core:

Adopting STM32H723VGT6 (550MHz), it meets the real-time requirements of Mecanum wheel kinematics calculation and multi-source data fusion.

Anti interference foundation:

Optimize power topology and communication bus, and build a robust hardware foundation.



Detailed plan and technical roadmap

Main control carrier board design

Core: STM32H723VGT6

Peripheral interface: Reserved UART/I2C/SPI/CAN, compatible with multiple sensors.

Clock and reset: high-precision crystal oscillator, hardware watchdog.

Power module

Input: Wide voltage DC-DC pre stabilization.

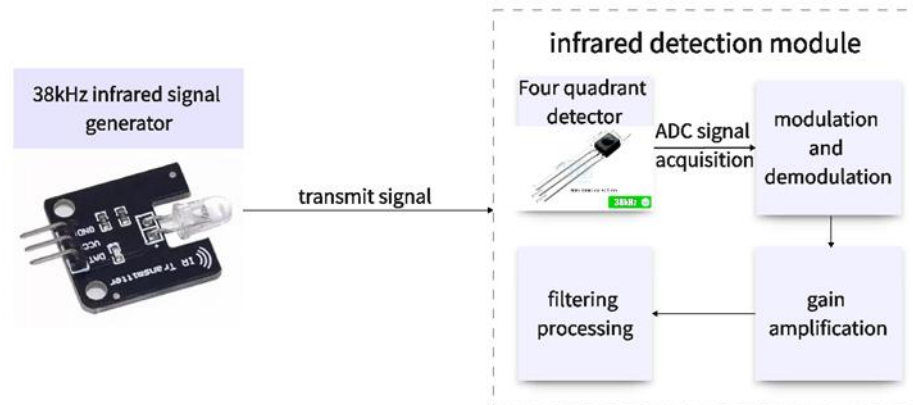
Output: Multiple independent LDOs provide power to MCU and sensors, with low noise.

Redundant design: Dual power hot backup to ensure operational stability.

Sensor circuit

360 ° ultrasonic radar: self-developed continuous rotating mechanism

Infrared locking array: high-frequency infrared transmitter/receiver



PCB layout and testing

Signal conditioning: operational amplifier amplification, comparator shaping, optocoupler isolation.

Thermal distribution simulation: layout optimization of key components (MOSFET, MCU), heat dissipation via holes.

Test plan: signal integrity test, ripple test, power load pull test, vibration test.

Innovation Points

- The single board integrates the main control, power supply, and interface, significantly reducing the size and minimizing interconnect latency.
- Achieving 360 ° dynamic environment perception, and improving tracking robustness.
- Achieves precise target locking during high-speed motion.
- Ensure uninterrupted power supply under extreme working conditions and improve system reliability.

Technical Challenge

- Heat dissipation management under high-density integration(Thermal distribution simulation and testing)
- Signal integrity assurance for 550MHz high-speed signals(Impedance control, crosstalk suppression)
- Hardware latency optimization for multi-sensor synchronization and data fusion
- Reliability of connectors and solder joints under vibration environment

Subtasks	Specific Content
3.1 Schematic Design	Design complete schematics for main carrier board using Altium Designer. Integrate STM32H723VGT6 MCU, power tree, sensor interfaces, and communication buses.
3.2 PCB Module Design	High-density PCB layout with impedance-controlled routing, thermal optimization, and power integrity design.
3.3 PCB Module Testing	Validate power rails (ripple <100mV), signal integrity, and communication protocols; perform thermal and vibration tests.
3.4 Carrier Board Design	Integrate PCB with motor and sensors; debug MCU–driver–sensor communication; optimize power design.

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[illegible]

Vision Perception Module

Lane Following Based on Threshold Segmentation

PIPELINE

Input image → ROI extraction → Thresholding → Lane center extraction → Error output

FORMULA

Binary segmentation

$$B(x, y) = H(I(x, y) - T)$$

where $H(\cdot)$ denotes thresholding and W_{ROI} is the ROI width.

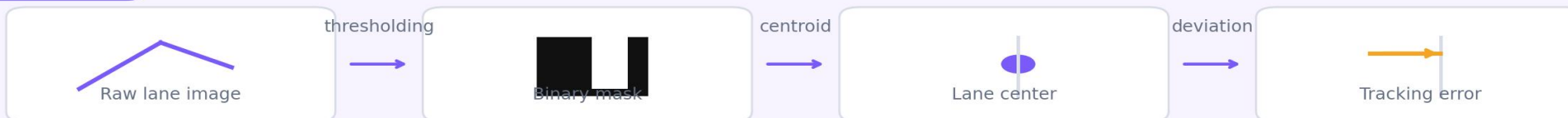
Lane center

$$x_c = \frac{\sum x B(x, y)}{\sum B(x, y)}$$

Normalized lateral error

$$e_y = \frac{x_c - x_{ref}}{W_{ROI}/2}$$

OUTPUT FLOW



Output: lane mask, lane center, and normalized lateral error for steering control

Vision Perception Module

Object Recognition Based on YOLO

PIPELINE

Input image → Backbone / Neck → Bounding box regression → NMS → Target output

FORMULA

$$\begin{aligned}b_x &= \sigma(t_x) + c_x, & b_y &= \sigma(t_y) + c_y \\b_w &= p_w e^{t_w}, & b_h &= p_h e^{t_h} \\s &= P(\text{obj}) \cdot P(\text{class} | \text{obj})\end{aligned}$$

OUTPUT FLOW



Output: target class, bounding box location, and confidence score

Subtasks	Specific Content
4.1 Training Set Collection	Calibrate the onboard camera and record video data under diverse lighting and road scenarios; extract frames and manually annotate lane boundaries for threshold-based detection and bounding boxes for YOLO training.
4.2 Model Training	Test each module independently to ensure that there are no issues with the modules during the later system development process.
4.3 Model Testing	Verify lane mask accuracy, center extraction, and lateral error under varied conditions while ensuring YOLO delivers correct bounding boxes and class scores with effective NMS; run both modules concurrently to assess real-time performance and format final outputs for control systems.

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AGV upper computer software

We have developed a **cross-platform, visual, and user-friendly** upper computer software using **Python**. Below are the **modules** we employed.



PySide6



We utilize it to create an **elegant GUI interface**. The GUI library it provides enabled the implementation of video windows, numerical/string input, and various buttons. It helped us achieve the **visualization** of the software, significantly **reducing the barrier** for users to use the AGV.

OpenCV



We utilize it to achieve **video stream retrieval and display**, and to retain the backend interface for the **tracking algorithm** for the vision teamer, allowing them to **input video and output mask images**.

Pyserial



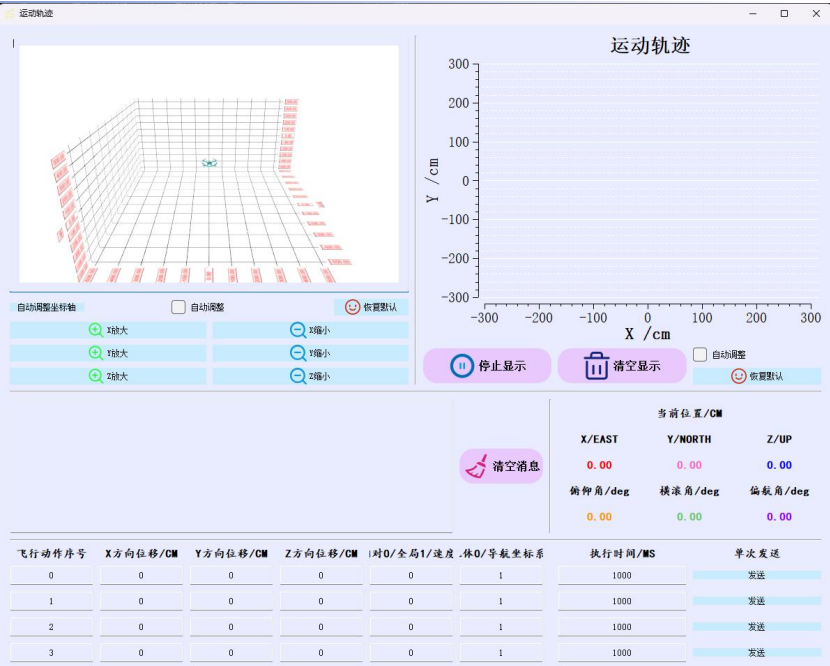
We use it to establish **serial communication** with the AGV, enabling the issuance of commands and the feedback of information.

AGV upper computer software

Pygam



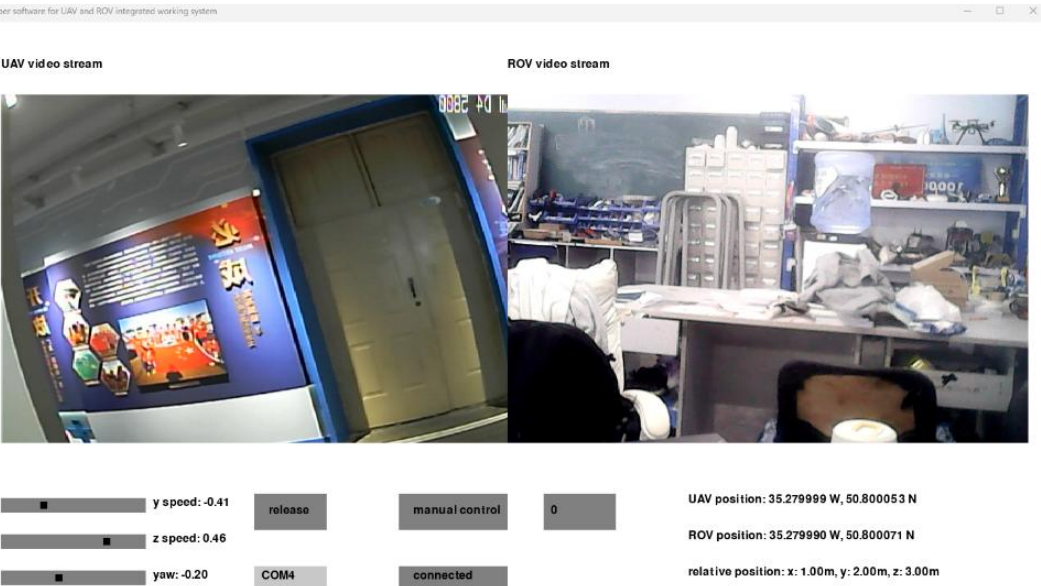
We utilize its **Joystick** module to implement the input for the game controller, enabling the software to read and process most of the controllers available on the market, making the operation of the AGV more **user-friendly**. It also utilize its animation library to develop the **visualization** of the obstacle avoidance system and navigation system, making our software more aesthetically pleasing.



PyTorch



We have utilize it to incorporate some **AI** models, such as **ultralytics'** YOLO26, enabling us to accomplish the task of **object detection** within the software. In the future, it could even be used as a **embodied intelligence platform**, further leveraging this module.



Subtasks	Specific Content
5.1 Backend Design	Develop various backend task modules, such as serial port communication, video stream retrieval, and each module adheres to the principle of low coupling.
2.2 Backend Testing	Test each module independently to ensure that there are no issues with the modules during the later system development process.
2.3 Frontend Design	Design the front-end GUI page, connect the back-end interface, and create a beautiful and elegant software.

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Budget Sheet

Category	Itemized Budget Details	Subtotal (CNY)	Percentage
Equipment Costs	1. Main Controller (STM32H723VGT6) and core electronic components procurement 2. Mecanum wheels, matching servo motors, and drive control modules procurement 3. Perception array (rotating ultrasonic radar, high-frequency infrared modules) procurement 4. High-rate power battery and power management topology components procurement	425	85%
Materials & Fabrication Costs	1. PCB prototyping and fabrication for the high-density chassis carrier board and sensor modules 2. Fabrication of vehicle structural components (3D-printed support parts, acrylic base plate) 3. Hardware connectors, fasteners, and specialized electrical wiring procurement	75	15%
Total Project Budget	—	500	—



Summary

This project developed a modular Mecanum-wheeled tracking robot , with its core innovation lying in the high integration of hardware and software. Mechanically, independent suspensions effectively solve the Mecanum wheel vibration issue. For control, we designed a custom high-density motherboard and built low-level algorithms like cascade PID, kinematics, and multitasking scheduling from scratch. Software-wise, a Python-based GUI integrated with the YOLO model enables precise visual perception and obstacle avoidance. The entire system operates stably and is fully controllable.

Looking ahead, we will further expand functions based on this stable platform. We plan to incorporate wireless communication modules to enable more flexible remote control. Furthermore, we will introduce machine learning algorithms for automatic parameter tuning , ultimately upgrading the robot into a powerful embodied intelligence platform.

Mecanum-wheeled Autonomous Tracking Robot Thesis Proposal

Thank you for watching

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汇报日期 2026. 3. 14

Trace Rover Team